

Qatar-2b

R-1948

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_130429 · created 2026-08-06T15:16:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456411.82891 +/- 0.00026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.04 +/- 0.033
Transit depth (Rp/Rs)^2	0.16 +/- 0.26 [%]
Orbital Inclination (inc)	84.12 +/- 1.7
Airmass coefficient 1 (a1)	0.871 +/- 0.026
Airmass coefficient 2 (a2)	0.091 +/- 0.023
Scatter in the residuals of the lightcurve fit is	2.99 %
Best Comparison Star	#2 - [523, 226]
Optimal Aperture	3.45
Optimal Annulus	13.12
Transit Duration (day)	0.05 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.04 ± 0.033 vs 0.16327 (deviation 3.54σ)
Duration fit vs published	0.05 d vs 0.0767 d (deviation 1.81σ)
Mid-time O-C	0.2 min
Depth SNR	1.1
Residual scatter	2.99 %

Light curve

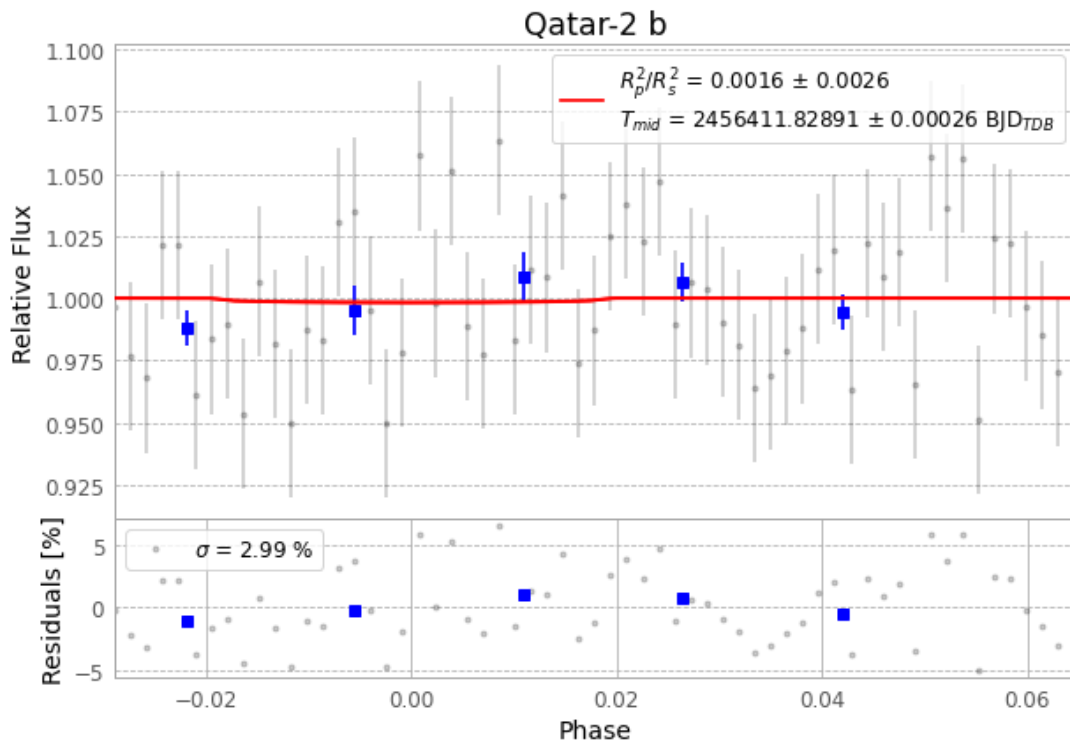


Figure 1 — FinalLightCurve_Qatar-2 b_2013-04-28.png (SHA-256 ed21e4579973d203...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 301], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 457e2086fc36d5c8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), Qatar-2130429064813.FITS → Qatar-2130429094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-130429.log (24ead1114e04...), FinalLightCurve_Qatar-2 b_2013-04-28.png (ed21e4579973...), FinalLightCurve_Qatar-2 b_2013-04-28.csv (7e69af11263e...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:16Z.